Supplementary Information

Supplementary Tables

Table S1: **Inverse temperature sensitivity analysis.** SA generation quality as a function of β/β^*, where $\beta^* \approx 2.94$ was identified via entropy inflection in the 18-dimensional PCA space (K=23 stored patterns). Operating at $\beta = \beta^*$ (bold row) balances generation novelty against fidelity. Med MRE = median relative error of per-feature means; Novelty = $1 - \max_k \cos(\hat{\xi}, \hat{m}_k)$.

β/β^*	β	Noise scale	PC1 std ratio	Novelty	Med MRE (%)
0.1	0.29	0.261	0.538	0.542	0.7
0.3	0.88	0.151	0.550	0.536	0.7
0.5	1.47	0.117	0.562	0.528	0.7
1.0	**2.94**	**0.082**	**0.589**	**0.502**	**0.6**
1.5	4.41	0.067	0.613	0.474	0.7
2.0	5.88	0.058	0.638	0.446	0.7
3.0	8.82	0.048	0.689	0.403	0.8

Table S2: **Multiplicity weighting parameters for conditional generation.** The multiplicity ratio ρ was computed to achieve a target effective fraction $f_{\text{target}} = 0.80$ of softmax attention on the designated patterns. K_{eff} is the participation ratio quantifying the effective number of patterns contributing to generation.

Condition	K_{sub}	K_{bg}	ρ	f_{eff}	K_{eff}
Uncomplicated	18	5	1.11	0.80	23.0
PCOS	3	20	26.67	0.80	4.6
Developed PE	5	18	14.40	0.80	7.7

Table S3: **SA generation hyperparameters.** The same values were used for unconditioned and conditioned generation except where noted.

Parameter	Value	Description
α	0.01	Langevin step size (see Table S8)
T	2,000	Langevin iterations per sample
β^*	2.94	Inverse temperature (entropy inflection)
PCA threshold	95%	Variance retained
d_{PCA}	18	PCA dimensionality
N	100	Synthetic patients per cohort
Random seed	42	For reproducibility
f_{target}	0.80	Target attention fraction (conditioned only)

Table S4: **BZ2012 ODE model calibration.** Five rate constants were calibrated on Visit 1 real patients (n=23) using bounded Nelder–Mead optimization in log-scale-factor space. The cost function was the mean normalized relative squared error across five TGA features under TF-only conditions. Bounds: $\pm\log(100)$. Convergence: $f_{\mathrm{tol}} = 10^{-6}$, 500 iterations, 11 random restarts.

Rate constant	Role	Default	Calibrated	Scale
prothrombinase k_{cat}	Thrombin burst	63.5 s^{-1}	21.94 s^{-1}	0.35×
intrinsic Xase k_{cat}	Amplification Xa	8.2 s^{-1}	0.169 s^{-1}	0.021×
extrinsic Xase k_{cat}	Initiation Xa	6.0 s^{-1}	93.2 s^{-1}	15.5×
PC activation k_{cat}	Protein C activation	0.41 s^{-1}	0.890 s^{-1}	2.17×
mIIa conversion k	mIIa→IIa	2.3×10^{8}	1.15×10^{9}	5.01×

Units for mIIa conversion: $\mathrm{M}^{-1}\mathrm{s}^{-1}$

Table S5: **Summary of mean relative errors across all 72 features.** Per-visit distribution of MREs between real and SA-generated synthetic patient means. The full feature-by-feature table (72 features × 3 visits = 216 entries) is provided in the code repository as `paper_summary_statistics.csv`.

Visit	Median MRE	Mean MRE	25th pctl	75th pctl	Max MRE	Below 5%
V1 (baseline)	0.022	0.029	0.008	0.039	0.158	59/72 (81.9%)
V2 (1st tri)	0.009	0.021	0.004	0.022	0.224	65/72 (90.3%)
V3 (3rd tri)	0.011	0.016	0.005	0.024	0.062	69/72 (95.8%)
Pooled (all 3 visits)	**0.012**	**0.022**	**0.005**	**0.028**	**0.224**	**193/216 (89.4%)**

Pooled median MRE 95% bootstrap CI: (0.98%, 1.60%) over 10,000 resamples (seed 42).

Table S6: **Comparison with deep generative baselines.** CTGAN and TVAE were trained on the same 90 per-visit records (23 patients $\times$ 3 visits, 72 features) at three epoch counts. SA and MVN results from the main text are shown for reference. CTGAN fails at this sample size (median MRE $\approx$ 19%), consistent with GAN mode collapse on small datasets. TVAE achieves comparable marginal fidelity at high epoch counts but cannot capture cross-visit covariance or perform conditional generation.

Method	Epochs	Med MRE	Med KS	Corr MAE
SA	–	0.019	0.169	0.200
MVN	–	0.024	0.161	0.090
CTGAN	300	0.189	0.364	0.261
CTGAN	1,000	0.195	0.349	0.265
CTGAN	3,000	0.187	0.341	0.180
TVAE	300	0.037	0.174	0.152
TVAE	1,000	0.026	0.248	0.082
TVAE	3,000	0.018	0.224	0.092

Table S7: **PCA variance threshold sensitivity analysis.** SA generation quality is stable across thresholds from 85% to 99%. Median MRE remains between 1.0% and 1.7% across all thresholds. The 95% threshold used in the main text (bold) is not a critical design choice.

Threshold	d_{PCA}	K/d	β^*	Novelty	Med MRE	Corr MAE
85%	13	1.77	2.94	0.420	0.0098	0.124
90%	15	1.53	2.94	0.477	0.0160	0.126
95%	**18**	**1.28**	**2.94**	**0.500**	**0.0124**	**0.143**
97.5%	20	1.15	2.94	0.543	0.0120	0.135
99%	21	1.10	2.94	0.541	0.0172	0.138

Table S8: **ULA vs MALA sampling diagnostics.** Ten chains of 5,000 iterations each were run with both the Unadjusted Langevin Algorithm (ULA) and the Metropolis-Adjusted Langevin Algorithm (MALA) at step size $\alpha = 0.01$ and $\beta^* = 2.94$ on the 18-dimensional PCA memory (K=23 patterns). The MALA acceptance rate of 99.9% confirms that virtually every ULA proposal satisfies detailed balance, and the indistinguishable energy and effective sample size statistics confirm negligible discretization bias.

Metric	ULA	MALA
Acceptance rate (%)	– (no rejection)	99.9 ± 0.03
Mean energy (post-burn-in)	1.932 ± 0.141	1.886 ± 0.231
τ_{int}	109.5 ± 49.5	106.7 ± 30.4
Effective sample size	41.8 ± 14.2	40.0 ± 10.2

Burn-in: 1,000 iterations discarded. τ_{int}: integrated autocorrelation time estimated with a windowed estimator (threshold 0.05). ESS = $(T - T_{\text{burn}})/\tau_{\text{int}}$.

Table S9: **Memorization diagnostics.** Two complementary checks that SA-generated patients are not memorized copies of the K=23 stored profiles. Novelty score is the angular distance between a synthetic sample and its nearest stored pattern at $\beta = \beta^*$. Nearest-neighbor distances are computed in the per-visit standardized 72-dimensional feature space.

Metric	Value
Mean novelty score, $1 - \max_k \cos(\hat{\xi}, \hat{m}_k)$	0.50
Median synth–real nearest-neighbor distance (std. units)	6.14
Median real–real nearest-neighbor distance (std. units)	6.87
Distance ratio (synth–real / real–real)	0.89

Table S10: **Bootstrap Mann–Whitney equivalence per feature and condition.** For each of 24 feature–condition pairs, we subsampled the synthetic cohort to match the real sample size, applied a Mann–Whitney U test, and repeated 1,000 times. Frac. $p > 0.05$ is the fraction of replicates in which the test could not distinguish the synthetic from the real distribution. 20 of 24 pairs achieve $p > 0.05$ in $\geq 90\%$ of replicates; the four failing pairs (italicized) all involve high-variance features in the smallest subgroups.

Condition	Feature	MRE	Frac. $p > 0.05$
Uncomplicated (n=18)	FVIII	0.057	0.952
Uncomplicated (n=18)	vWF	0.022	0.999
Uncomplicated (n=18)	FIX	0.018	0.998
Uncomplicated (n=18)	FV	0.006	0.996
Uncomplicated (n=18)	α_2-AP	0.019	1.000
Uncomplicated (n=18)	FVII	0.057	0.941
Uncomplicated (n=18)	Fbgn	0.017	0.987
Uncomplicated (n=18)	Plgn	0.020	0.988
PCOS (n=3)	FVIII	0.109	0.985
PCOS (n=3)	vWF	0.150	0.959
PCOS (n=3)	*FIX*	*0.121*	*0.873*
PCOS (n=3)	FV	0.057	0.944
PCOS (n=3)	α_2-AP	0.051	0.987
PCOS (n=3)	FVII	0.029	0.999
PCOS (n=3)	Fbgn	0.043	0.992
PCOS (n=3)	*Plgn*	*0.153*	*0.615*
Developed PE (n=5)	FVIII	0.097	0.932
Developed PE (n=5)	vWF	0.010	0.986
Developed PE (n=5)	*FIX*	*0.108*	*0.780*
Developed PE (n=5)	FV	0.006	0.987
Developed PE (n=5)	α_2-AP	0.062	0.950
Developed PE (n=5)	FVII	0.045	0.993
Developed PE (n=5)	Fbgn	0.048	0.996
Developed PE (n=5)	*Plgn*	*0.093*	*0.814*

Median Frac. $p > 0.05$ across all 24 pairs: 0.986. Failing pairs (< 0.90, italicized): PCOS–FIX, PCOS–Plgn, DPE–FIX, DPE–Plgn.

Table S11: **Mechanistic validation diagnostics per feature and condition.** Cloud overlap is the fraction of synthetic ODE-predicted/measured ratios falling within the 5th–95th percentile of the corresponding real distribution. KS D and KS p are from a two-sample Kolmogorov–Smirnov test on the same ratio distributions. Real n=69 (23 patients × 3 visits); synthetic n=300 (100 patients × 3 visits).

Condition	TGA Feature	Cloud overlap	KS D	KS p
TF-only	Lagtime	0.92	0.112	0.46
TF-only	Peak	0.91	0.081	0.84
TF-only	T.Peak	0.93	0.102	0.58
TF-only	Max Rate	0.92	0.083	0.82
TF-only	ETP	0.86	0.123	0.34
TF+TM	Lagtime	0.93	0.127	0.30
TF+TM	Peak	0.91	0.079	0.87
TF+TM	T.Peak	0.93	0.110	0.49
TF+TM	Max Rate	0.89	0.096	0.65
TF+TM	ETP	0.93	0.112	0.46

TF-only cloud overlap range: 0.86–0.93. TF+TM cloud overlap range: 0.89–0.93. KS D range across both conditions: 0.079–0.127, all $p > 0.30$.

Table S12: **Downstream utility — per-feature held-out V2/V3 performance.** Median relative error of the BZ2012 ODE-predicted TGA values against the held-out real V2 and V3 measurements, for the model calibrated on real V1 patients (K=23) and the model calibrated on synthetic V1 patients (N=100). The synth-calibrated model achieves slightly lower error on every feature. Synth/Real ratio < 1 favors the synth-calibrated model.

TGA Feature	Real-cal MRE	Synth-cal MRE	Synth/Real
Lagtime	0.165	0.162	0.98×
Peak	0.097	0.087	0.90×
T.Peak	0.330	0.306	0.93×
Max Rate	0.286	0.259	0.91×
ETP	0.197	0.183	0.93×
Overall median	**0.201**	**0.189**	**0.94×**

Supplementary Figures

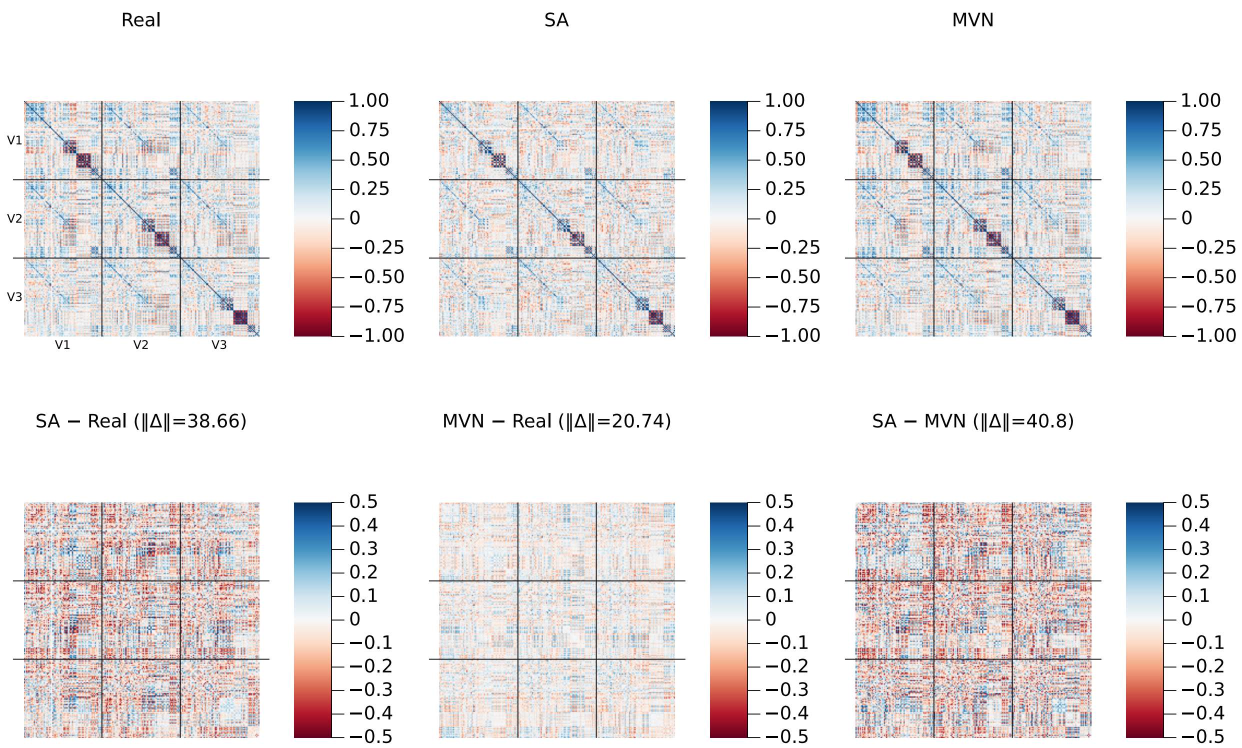

Figure S1: **Cross-visit correlation structure (amplified residual scale).** Same layout as the main-text correlation figure, but with the bottom-row residual panels plotted on a ± 0.5 color scale (vs. ± 1.0 in the main text) to reveal fine structure. Frobenius norms of each residual matrix are shown in the panel titles.

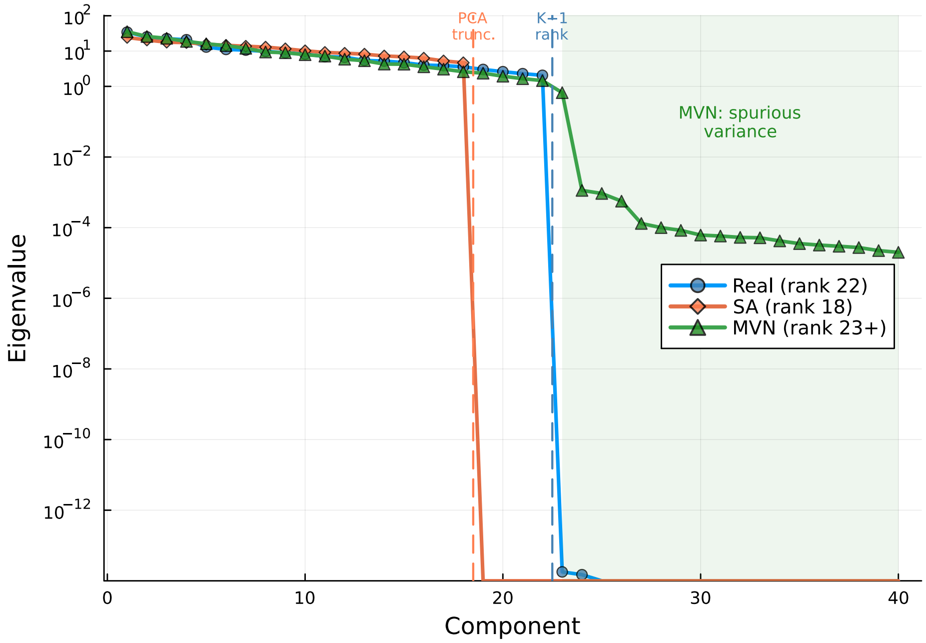

Figure S2: **Covariance eigenvalue spectrum.** Log-scale eigenvalues of the 216×216 sample covariance for Real (K=23, rank 22), SA (N=100, rank 18), and MVN (N=100) populations. Dashed vertical lines mark the SA PCA truncation point (component 18) and the data rank limit (component 22). MVN's Ledoit–Wolf regularization inflates eigenvalues beyond component 22 (green shaded region), introducing spurious variance in 194 dimensions where the data contain no signal.

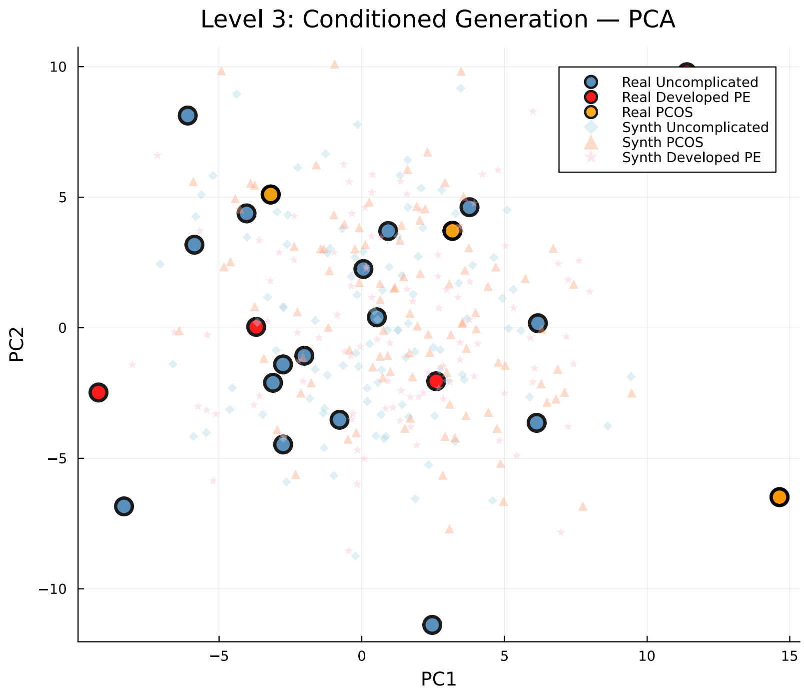

Figure S3: **Conditioned generation: PCA projection (pooled).** Large markers: real patients (blue = Uncomplicated, n=18; orange = PCOS, n=3; red = Developed PE, n=5). Small markers: synthetic patients (N=100 per condition). With only 3 PCOS and 5 Developed PE patients, the subgroups do not form clearly separable clusters in the first two principal components, but the synthetic cohorts concentrate around their respective real counterparts.

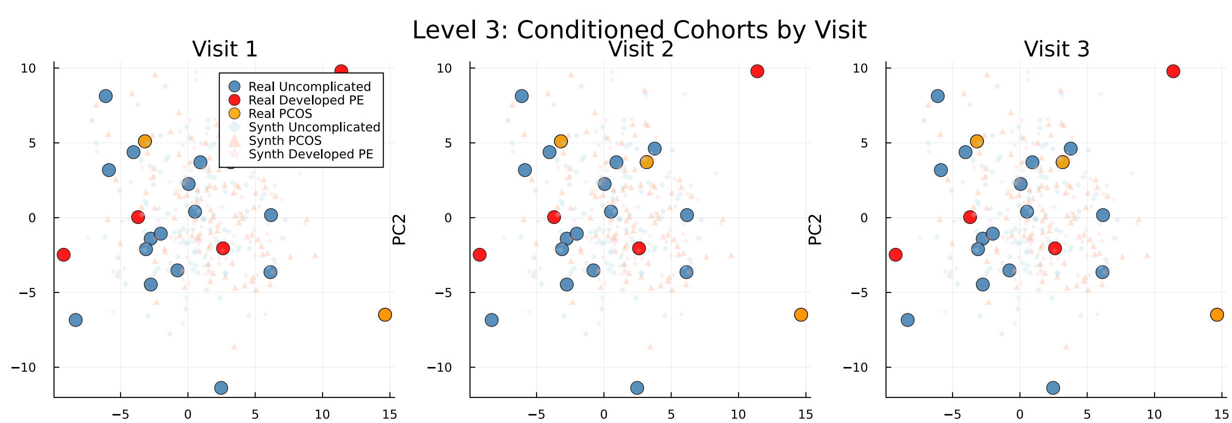

Figure S4: **Conditioned generation: PCA projection by visit.** Same as Supplementary Fig. S3 but separated by visit (V1, V2, V3). Condition-specific clustering is maintained within each individual visit.

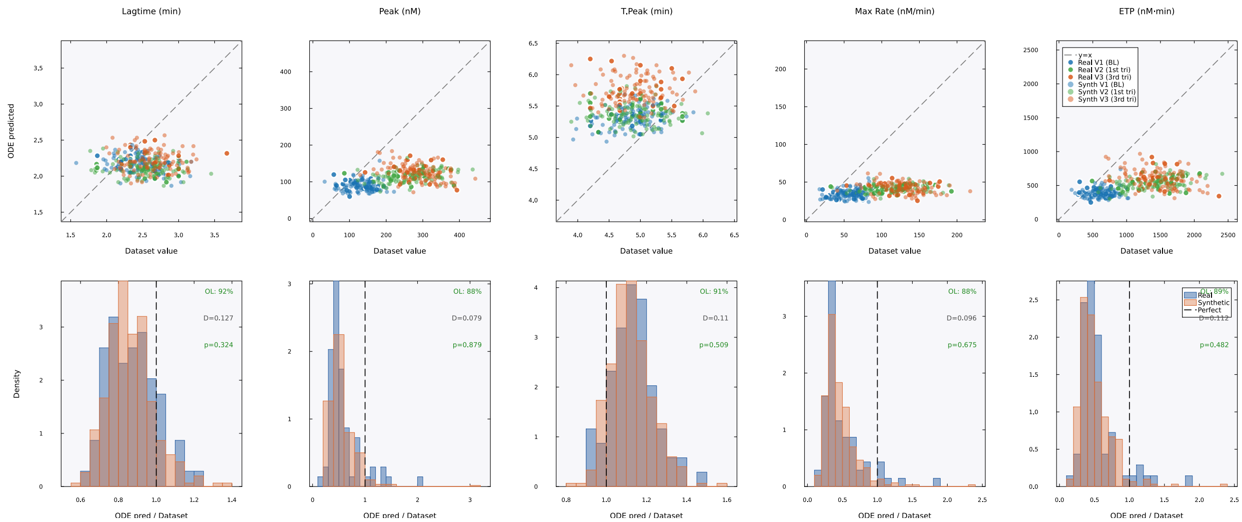

Figure S5: **Mechanistic validation (TF+TM condition).** Same layout as main-text Fig. 6 but under TF+TM conditions. The model systematically underpredicts peak and maximum rate ($\approx 0.5\times$ measured) for both real and synthetic patients, reflecting overcorrection of the protein C anticoagulant pathway. Cloud overlap remains high at 88–92%, confirming that the model processes real and synthetic patients identically even under imperfect calibration conditions.

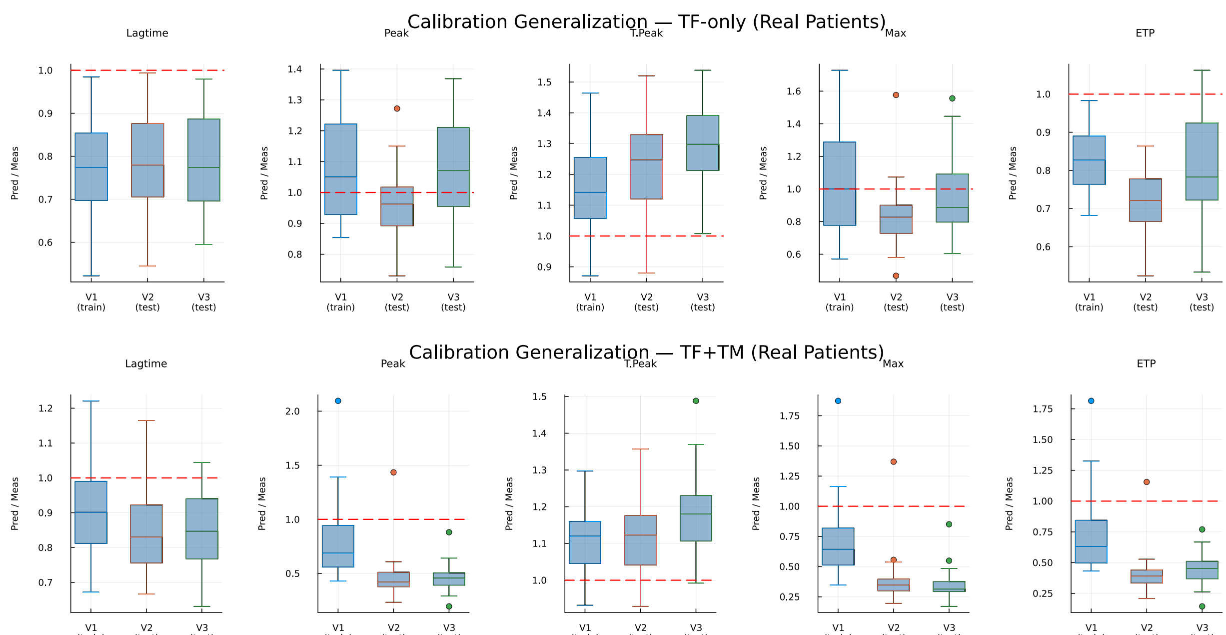

Figure S6: **Cross-visit calibration generalization.** Boxplots of the ODE-predicted/dataset ratio by visit for real patients. The model was calibrated on Visit 1 only. Top: TF-only conditions, where ratios near 1.0 at Visits 2 and 3 indicate that the calibration generalizes across pregnancy timepoints. Bottom: TF+TM conditions, where the calibration does not generalize as well, particularly for peak and maximum rate.

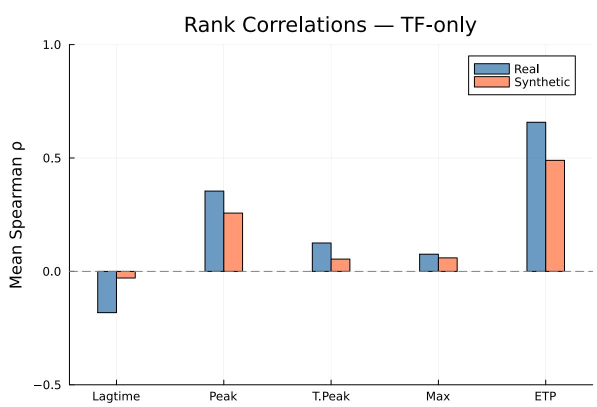

Figure S7: **Spearman rank correlations between ODE-predicted and dataset TGA features.** Left: TF-only. Right: TF+TM. ETP is the best-predicted feature ($\rho \approx 0.6$–0.8 for real patients under TF-only). Weak rank correlations for other features are expected given the 5-parameter population-level calibration.